Aleš Spital

Software Engineer (Unity/C#) | Multiplayer Systems | AI and Pathfinding | Performance | Tools and Data-Driven Gameplay

Velenje, Slovenia (EU) | Freelance/Contract - Remote (European time zones) | Mobile: +386 31 314 310
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SUMMARY

Unity/C# software engineer (M.Sc. Computer Science) delivering performance-sensitive, interactive products across standalone VR (Meta Quest), AR, and mobile-first prototypes. I build clean, modular systems (thin MonoBehaviours, most logic in plain C#), profile early, and harden features with reproducible debugging and defensive invariants. Seeking remote freelance/contract collaboration on sandbox/simulation gameplay with a strong engineering culture.

CORE SKILLS

- Unity and C#: modular services and systems, ScriptableObjects for data-driven content, UI (uGUI, TextMeshPro), physics and animation, URP familiarity.
- Multiplayer: Netcode for GameObjects patterns (RPCs, replicated state, ownership transfer), server-authoritative interaction workflows.
- AI and pathfinding: NavMesh workflows, steering and obstacle avoidance, simple flocking; behavior-state patterns; A* concepts.
- Performance: Unity Profiler, allocation/GC control, frame-time stability; mobile/Quest constraints; Jobs/Burst when it pays off.
- Code culture fit: plain C# over heavy MonoBehaviour logic, explicit lifetimes/cleanup rules, guard clauses and assert-style invariants; comfortable with C# nullable reference types; test-minded checks where applicable.
- Collaboration: Git workflows, readable diffs, disciplined bug isolation, regression checklists, clear handoffs.

EXPERIENCE

Lead Software Developer (Unity/XR) - VR4LL 2.0 (EU-funded Erasmus+)

Tehnološki park Ljubljana (TPLJ) | 2023-2025

- Led development of three standalone Meta Quest environments from prototype to release-ready builds, prioritizing usability, stability, and interaction feel.
- Built multiplayer interaction and progression flows (replicated object interaction and session state), plus voice chat integration for multi-user sessions.
- Profiled and optimized under standalone headset constraints: reduced allocations, tightened hot paths, and stabilized frame-time in interaction loops.
- Debugged cross-device and build issues using reproducible test scenes, defensive sanity checks, and regression routines.

C#/.NET Developer (Xamarin.Forms)

Mega M d.o.o., Slovenia | 2019-2020

- Built core app architecture and primary UI flows (login, navigation, data presentation) for a production business app.
- Integrated REST APIs; handled JSON/XML payloads; collaborated via Git; worked with MySQL-backed data.

Additional experience: AI Workshop Instructor and Advisor (2025) | High School Professor - Software/IT (2021-2024) | Software Developer Intern (2018-2019)

SELECTED PROJECTS

- RyftRealm (in development): mobile-first Unity sandbox/RPG with data-driven inventory and economy (sell rules, shop rules, loot bundles); editor tooling for content authoring; performance-first architecture with future server authority in mind.
- ARnet (Google Play): AR app for building and exploring networking simulations with interactive 3D content and guided lessons. play.google.com/store/apps/details?id=com.CriticalGlitch.ARnet

EDUCATION AND LANGUAGES

- M.Sc. Computer Science and IT - University of Maribor (2020-2024) | B.Sc. (2016-2020)
- Languages: Slovenian (native) | English (fluent)